

BUMET 1MG TABLET

Composition

Each tablet contains Bumetanide 1mg

Product Description

A white to off white round shaped tablet with breaking line on one side

Pharmacodynamics

BUMET is a potent, high ceiling loop diuretic with a rapid onset and a short duration of action. The primary site of action is the ascending limb of the Loop of Henlé where it exerts inhibiting effects on electrolyte reabsorption causing the diuretic and natriuretic action observed.

Pharmacokinetic

BUMET is well absorbed after oral administration with the bioavailability reaching between 80 and 95%. The elimination half-life ranges from between 0.75 to 2.6 hours. No active metabolites are known. Renal excretion accounts for approximately half the clearance with hepatic excretion responsible for the other half. There is an increase in half-life and a reduced plasma clearance in the presence of renal or hepatic disease. In patients with chronic renal failure the liver takes more importance as an excretory pathway although the duration of action is not markedly prolonged.

The mode of action has been determined through various clearance studies in both humans and experimental animals. BUMET inhibits sodium reabsorption in the ascending limb of the loop of Henle, as shown by marked reduction of free-water clearance (CH_2O) during hydration and tubular free-water reabsorption (TCH_2O) during hydropenia. Reabsorption of chloride in the ascending limb is also blocked by BUMET, and BUMET is somewhat more chloruretic than natriuretic.

Potassium excretion is also increased by BUMET, in a dose-related fashion.

BUMET may have an additional action in the proximal tubule. Since phosphate reabsorption takes place largely in the proximal tubule, phosphaturia during BUMET induced diuresis is indicative of this additional action. This is further supported by the reduction in the renal clearance of BUMET by probenecid, associated with diminution in the natriuretic response. This proximal tubular activity does not seem to be related to an inhibition of carbonic anhydrase. BUMET does not appear to have a noticeable action on the distal tubule.

BUMET decreases uric acid excretion and increases serum uric acid. Following oral administration of BUMET the onset of diuresis occurs in 30 to 60 minutes. Peak activity is reached between 1 and 2 hours. At usual doses (1 mg to 2 mg) diuresis is largely complete within 4 hours; with higher doses, the diuretic action lasts for 4 to 6 hours.

Indication

BUMET is indicated whenever diuretic therapy is required in the treatment of oedema e.g. associated with congestive heart failure, cirrhosis of the liver hepatic, renal disease, including the nephrotic syndrome.

Acute pulmonary oedema, drug-induced fluid retention and drug poisoning that can be treated by forced diuresis. Hypertension.

Recommended Dose

Orally : 1mg
daily

In refractory cases the dose can be increased gradually till a satisfactory response has been obtained. Rarely will it be necessary to exceed a dose of 4mg a daily.

Children: The dose is calculated on the basis of 0.03-0.06 mg/kg daily.

Elderly: Adjust dosage according to response; a dose of 0.5mg daily may be sufficient in some elderly patients.

Route of Administration:

Oral Administration

Contraindication

BUMET is contraindicated in anuria. Although BUMET can be used to induce diuresis in renal insufficiency, any marked increase in blood urea nitrogen or creatinine, or the development of oliguria during therapy of patients with progressive renal disease, is an indication for discontinuation of treatment with BUMET. BUMET is also contraindicated in patients in hepatic coma or in states of severe electrolyte depletion until the condition is improved or corrected. BUMET is contraindicated in patients hypersensitive to this drug

Warnings and Precautions

The precautions to be taken with BUMET are mainly those associated with electrolyte disturbance. Electrolyte depletion symptoms are weakness, dizziness, lethargy, leg cramps, anorexia, vomiting or mental confusion. Patients in whom a risk of depletion is likely to undergo periodic serum electrolyte determination.

BUMET increases the excretion of potassium. This may cause the gradual development of low serum potassium levels. Patients on long term treatment should, therefore, be encouraged to take a high potassium diet. Potassium chloride supplements are indicated in those patients whose dietary potassium is possibly inadequate, the chloride tending to correct the hypochloraemia and metabolic alkalosis, which is occasionally associated with potassium depletion. Potassium sparing diuretics, such as spironolactone, have been used as an alternative approach. Supplemental potassium and/or spironolactone may prevent hypokalemia and metabolic alkalosis in patients with hepatic cirrhosis and ascites. Low serum potassium levels, it should be noted, increase the sensitivity of the myocardium to the toxic effects of digitalis. Serum potassium should be measured periodically and potassium supplements or potassium-sparing diuretics added if necessary. Periodic determinations of other electrolytes are advised in patients treated with high doses or for prolonged periods, particularly in those on low-salt diets.

It is also important to prevent hypokalaemia in patients with hepatic cirrhosis. Potassium supplements are also indicated in conditions associated with a particular tendency to potassium depletion, eg. long-term treatment with corticosteroids, ulcerative colitis, prolonged vomiting and diarrhea. For high dose therapy, treatment should be initiated at a low dose and gradually increased in 5mg increments until the desired response is obtained.

Patients allergic to sulfonamides may show hypersensitivity to BUMET.

Since there have been rare spontaneous reports of thrombocytopenia, patients should be observed regularly for possible occurrence of thrombocytopenia.

Hyperuricemia may occur; it has been asymptomatic in cases reported to date. Reversible elevations of the BUN and creatinine may also occur, especially in association with dehydration and particularly in patients with renal insufficiency. BUMET may increase urinary calcium excretion with resultant hypocalcemia.

Diuretics have been shown to increase the urinary excretion of magnesium; this may result in hypomagnesemia.

Effects on ability to Drive and Use Machine

BUMET has no or negligible direct influence on the ability to drive and use machines. However, the patient should be informed that dizziness may occur during treatment and take this into account while driving or using machines.

Interactions with other medicaments

Bumetanide shows a tendency to increase the excretion of potassium which can lead to an increase in the sensitivity of the myocardium to the toxic effects of digitalis. Thus the dose may need adjustment when given in conjunction with cardiac glycosides.

Bumetanide may potentiate the effects of antihypertensive drugs. Therefore, the dose of the latter may need adjustment when bumetanide is used to treat oedema in hypertensive patients.

Bumetanide may cause an increase in blood uric acid. Periodic checks on urine and blood glucose should be made in diabetics and patients suspected of latent diabetes.

Certain non-steroidal anti-inflammatory drugs have been shown to antagonise the action of diuretics.

Pregnancy and Lactation

In the ordinary precaution, the usage of BUMET must be avoided in the first trimester of pregnancy should at present be observed. Since it is not known whether bumetanide is distributed into breast milk, nursing should not be undertaken.

Undesirable Effects

Reported reactions include abdominal pain, vomiting, dyspepsia, diarrhoea, stomach and muscle cramps, arthralgia, dizziness, fatigue, hypotension, headache, nausea, encephalopathy (in patients with pre-existing hepatic disease), fluid and electrolyte depletion, dehydration, hyperuricaemia, raised blood urea and serum creatinine, hyperglycaemia, abnormalities of serum levels of hepatic enzymes, skin rashes, pruritus, urticaria, thrombocytopenia, gynaecomastia and painful breasts. Bone marrow depression associated with the use of BUMET has been reported rarely but it has not been proven definitely to be attributed to the drug. Hearing disturbance after administration of BUMET is rare and reversible.

Overdose and treatment

Symptoms would be those caused by excessive diuresis. Empty stomach by gastric lavage or emesis. General measures should be taken to restore blood volume, maintain blood pressure and correct electrolyte disturbance.

Presentation

10 tablets in a blister. Box of 10, 30, 60 or 100 tablets.

Product Registration Holder and Manufacturer

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